Problem Set 2

For some of these problems, you may want to use the visualization for S_3 and S_4 available at gtbarkley.org/cmnd/.

1. Verify that, in the symmetric group S_4, the elements $s_2s_1s_3s_2$ and $s_2s_1s_2$ are incomparable in weak order.

2. Prove that weak order on a Coxeter group is a partial order.

3. Prove from the definition of weak order that if $u \lessdot_R v$ is a cover relation, then $\ell(u) = \ell(v) - 1$ and there exists some $s \in S$ so that $us = v$.

4. Draw the Hasse diagram for weak order on the dihedral group $I_2(4)$.

5. Let W be a Coxeter group of rank n. Show that for the weak order on W, the number of elements that w covers plus the number of elements that cover w is equal to n.

6. We say that a permutation π has a 312**-pattern** if there exist $a < b < c$ so that $\pi^{-1}(c) < \pi^{-1}(a) < \pi^{-1}(b)$. (This just means that the three numbers appear in the order "biggest then smallest then middle".) We say that π is 312**-avoiding** if it has no 312 patterns.
   (a) How many 312-avoiding permutations are there in S_3? S_4? (Bonus problem: How many in S_n?)
   (b) Let X_n be the set of 312-avoiding permutations in S_n. The weak order turns X_n into a poset called the **Tamari lattice**. Draw the Tamari lattice X_3. (Time-consuming bonus: draw X_4.)

   The Tamari lattice X_n is related to cluster algebras. There is a bijection between X_n and the seeds of the type A_{n-1} cluster algebra. The graph you get by drawing an edge between two seeds related by a mutation turns out to be the same as the Hasse diagram for X_n. This graph also appears as the edge graph of a polytope called the **associahedron**. You can actually use the weak order on other Coxeter groups to understand other (finite-type) cluster algebras in a similar way, using **Cambrian lattices**.

7. Describe when two permutations of S_4 are on the same face of the simplex model for the Coxeter complex. Describe when two permutations of S_4 are on the same face of the permutahedron. What do you think the generalization is to S_n?

8. Using the braid arrangement model for S_4, verify that if a straight line segment is drawn between the permutation w and the identity permutation, then the planes that it passes through are exactly the inversions of w.

9. In this problem, assume that W is a Coxeter group with finitely many elements.
   (a) Show that there is an element $w_0 \in W$ such that, for all $w \in W$, we have $w \leq_R w_0$.
   (b) Show that, for all $w \in W$, the inversion set $\mathrm{inv}(ww_0) = T \setminus \mathrm{inv}(w)$.
   (c) Show that if $w \in W$ is an element such that $S \subseteq \mathrm{inv}(w)$, then $w = w_0$.
   (d) Since every element of W is a prefix of some reduced word of w_0, there must be quite a few reduced words for w_0. Find three different reduced words for w_0 in S_4.

10. The following problems are about inversions in the symmetric group S_n.
    (a) Prove that if π is a permutation, then the inversion set of π has the following two properties
        (i) If $1 \leq a < b < c \leq n$, and both transpositions (a, b) and (b, c) are in $\mathrm{inv}(\pi)$, then $(a, c) \in \mathrm{inv}(\pi)$.
        (ii) If $1 \leq a < b < c \leq n$, and neither transposition (a, b) nor (b, c) is in $\mathrm{inv}(\pi)$, then (a, c) is not in $\mathrm{inv}(\pi)$.
    (b) (Optional) Prove that if X is a set of transpositions satisfying properties (i) and (ii), then there is a permutation π with $\mathrm{inv}(\pi) = X$.

11. Prove the following properties of a Coxeter group W (if you didn't already yesterday)
    (a) Strong Exchange Property: If $t \in \mathrm{inv}(w)$ and $s_{i_1} \cdots s_{i_k}$ is a reduced word for w, then there exists a j so that $tw = s_{i_1} \cdots \widehat{s}_{i_j} \cdots s_{i_k}$. (The hat notation means to omit s_{i_j} from the word.)
    (b) Exchange Property: If s is a descent of w and $s_{i_1} \cdots s_{i_k}$ is a reduced word for w, then there exists a j so that $ws = s_{i_1} \cdots \widehat{s}_{i_j} \cdots s_{i_k}$.
    (c) Deletion Property: If $s_{i_1} \cdots s_{i_k}$ is a non-reduced word for w, then there exist j and j' so that $w = s_{i_1} \cdots \widehat{s}_{i_j} \cdots \widehat{s}_{i_{j'}} \cdots s_{i_k}$.

12. Define the support of the word $s_{i_1} \cdots s_{i_k}$ to be the set $\{s_{i_1}, \ldots, s_{i_k}\}$. Show that any two reduced words for an element w in a Coxeter group W have the same support.

13. Prove that the weak order on the symmetric group S_n is a **lattice**: this means that there is a least upper bound and greatest lower bound for any $x, y \in S_n$. You can just show there are least upper bounds. A least upper bound, or **join**, for x and y is an element z_0 in $Z = \{z \in S_n \mid x \leq_R z \text{ and } y \leq_R z\}$ such that for any $z \in Z$, we have $z_0 \leq_R z$. (Bonus problem: prove the lattice property for weak order on any finite Coxeter group.)

14. There are two simple ways to transform a reduced word in S_n into a different reduced word. One is to use the fact that $s_{i_j} s_{i_{j+1}} = s_{i_{j+1}} s_{i_j}$ whenever s_{i_j} and $s_{i_{j+1}}$ commute, allowing us to swap those two simple reflections in the reduced word without changing its product. This is called a commutation move. The other is to use the fact that $s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1}$, which allows us to replace a string of three consecutive letters matching the pattern $s_i s_{i+1} s_i$ or $s_{i+1} s_i s_{i+1}$ with the other pattern. This is called a braid move. Prove that any two reduced words for $w \in S_n$ can be transformed into each other by performing a sequence of commutation and braid moves. (This is called Matsumoto's Theorem.)